

End-to-End Data Management Pipeline for a Recommendation System

RecoMart — E-Commerce Recommendation Engine

Data Management for Machine Learning (AIMLCZG529 / DSECLZG529)

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1. Problem Statement

RecoMart is an e-commerce startup that wants to build a data-driven product recommendation engine to improve customer engagement and cross-selling opportunities. The platform collects user behaviour data from multiple channels — web/mobile clickstream logs, transactional purchase history, product catalog metadata, and external API data.

The goal of this project is to design and implement a complete, automated, modular data management pipeline that continuously ingests raw data, validates and cleans it, engineers features, trains recommendation models, and tracks every artefact along the way — following the practices of modern ML data stacks.

The business objective is to increase conversion rates by providing personalized product recommendations to users based on their browsing behaviour, purchase history, and explicit ratings.

2. Objectives

1. Ingest data from heterogeneous sources (CSV files + REST APIs) with error handling, retry logic, and audit logging.
2. Store raw data in a structured, partitioned data lake layout.
3. Profile and validate data quality (missing values, duplicates, schema mismatches, range violations).
4. Clean and preprocess data — handle missing values, encode categoricals, normalise numerics.
5. Engineer user-level, item-level, and user-item interaction features suitable for collaborative and content-based filtering.
6. Manage features through a versioned feature store with a metadata registry.
7. Version all datasets (raw, processed, features) with SHA-256 integrity tracking and a manifest.
8. Train and evaluate recommendation models (SVD collaborative filtering + content-based cosine similarity).
9. Track experiments, parameters, and metrics with MLflow.
10. Orchestrate the full pipeline as a DAG using Prefect (with a sequential fallback).

3. Data Sources

The pipeline ingests data from multiple heterogeneous sources, simulating a real-world e-commerce environment. Synthetic data is generated with intentional quality issues (missing values, duplicates, out-of-range ratings) to demonstrate the validation and cleaning pipeline.

Source	Type	Records	Description
Users	CSV	1,000	Demographics — age, gender, country, signup date, premium status
Products	CSV + JSON (API)	500	Catalog — name, category, price, brand, avg rating
Clickstream	CSV	~51,000	Events — views, clicks, add-to-cart, wishlist, search
Transactions	CSV	10,000	Purchases — quantities, amounts, payment methods
Ratings	CSV	~20,000	Explicit user-item ratings (1–5 scale) with review text
External API	REST (FakeStore)	20	Supplementary product data from live API

4. Methodology / Pipeline Architecture

The pipeline follows a modular, stage-based architecture where each component operates independently and produces structured outputs consumed by the next stage. The full DAG is:

Data Generation → Data Ingestion → Data Validation → Data Preparation → Feature Engineering → Feature Store → Data Versioning → Model Training

All stages are orchestrated by Prefect (with a sequential fallback) and produce timestamped logs and JSON reports for auditability.

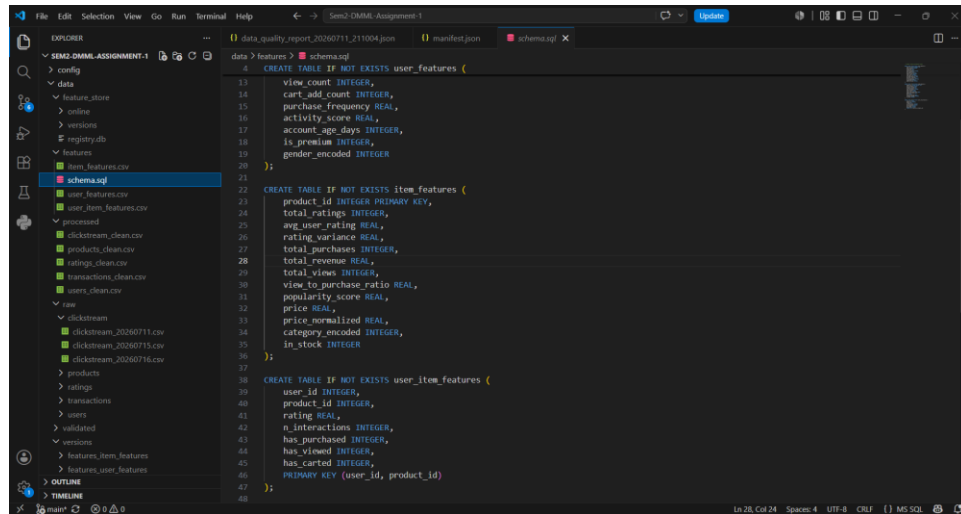


Figure 1: Data Lake Storage Structure

5. Implementation Details

5.1 Data Generation & Ingestion

Stage 1 generates realistic synthetic e-commerce data using the Faker library with fixed seeds for reproducibility. It intentionally injects ~5% missing values in user ages, ~3% in genders, ~2% missing transaction amounts, ~2% duplicate clickstream events, and ~1% out-of-range ratings.

Stage 2 ingests data from two source types: CSV files read from the local filesystem, and product data fetched from the FakeStore REST API (<https://fakestoreapi.com/products>). The API ingestion uses exponential-backoff retry (up to 3 attempts) with a 30-second timeout. All ingestion events are logged with Loguru and a JSON ingestion report is saved per run.

Key files: `src/ingestion/generate_data.py`, `src/ingestion/ingest.py`

5.2 Raw Data Storage

Ingested data is stored in a local data lake under `data/raw/`, structured as:

`data/raw/{source_name}/{source}_{YYYYMMDD}.csv` — partitioned by source and date-stamped filenames. This mirrors production data lake patterns (e.g., S3 prefix-based partitioning).

Storage structure is documented in the project README and `pipeline_config.yaml`.

5.3 Data Validation & Quality Report

The validation module performs four categories of checks on every dataset:

- Schema validation: Verifies required columns exist and detects extra/missing columns.
- Missing value analysis: Per-column and overall missing percentages.
- Duplicate detection: Row-level duplicate counts and percentages.
- Range validation: Rating values must be in [1–5], prices non-negative, quantities positive.

A comprehensive JSON quality report is generated per run and saved to `reports/`. Validated copies of each dataset are saved to `data/validated/`.

Key file: `src/validation/validate.py`

```
606  "ratings": {
607  },
608  "schema": {
609    "missing_columns": [],
610    "extra_columns": [],
611    "column_count": 5,
612    "expected_count": 5,
613    "issues": []
614  },
615  "missing_values": {
616    "total_missing_cells": 5848,
617    "total_missing_pct": 5.97,
618    "per_column": {
619      "review_text": {
620        "count": 5848,
621        "pct": 29.84
622      }
623    },
624    "duplicates": {
625      "total_duplicates": 0,
626      "duplicate_pct": 0.0
627    },
628    "range_checks": {
629      "range_issues": [
630        "Ratings out of range [1-5]: 193 rows"
631      ],
632      "n_issues": 1
633    },
634    "quality_status": "WARN",
635    "quality_issues": [
636      "Range violations"
637    ]
638  }
639 }
```

Figure 2: Data Quality Validation Report

5.4 Data Preparation & Exploratory Data Analysis

The preparation module performs the following cleaning and preprocessing steps:

- Missing value imputation: Median for ages, recalculated for transaction amounts, 'Unknown' for categoricals.
- Categorical encoding: LabelEncoder for gender, product category, event type, device, payment method.
- Numerical normalisation: MinMaxScaler for product prices.
- Derived features: Account age in days, hour-of-day and day-of-week from timestamps.
- Filtering: Removes out-of-range ratings (< 1 or > 5), deduplicates clickstream events.

Key file: `src/preparation/prepare.py`, `notebooks/eda_analysis.ipynb`

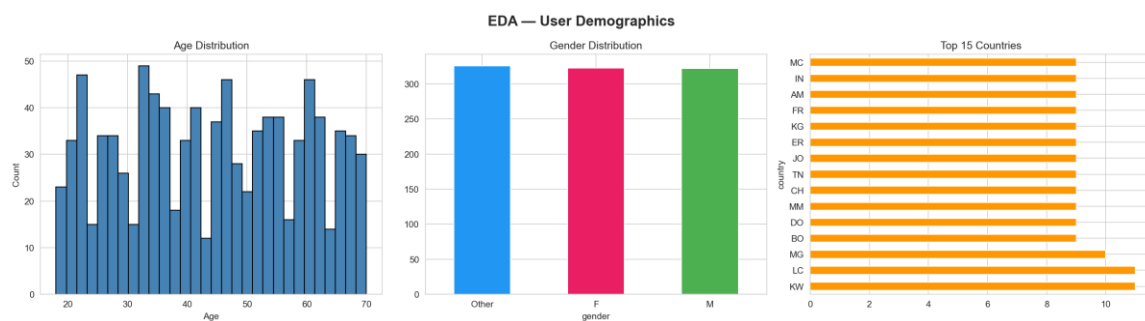


Figure 3: User Demographics — Age, Gender, Country Distribution

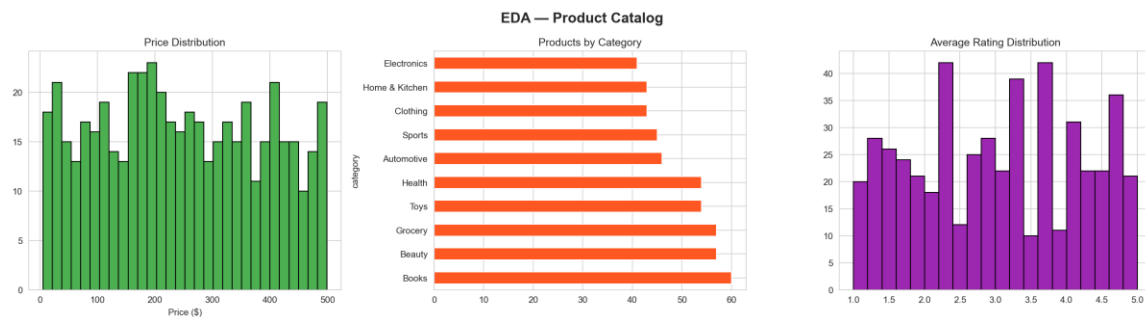


Figure 4: Product Catalog — Price, Category, Rating Distribution

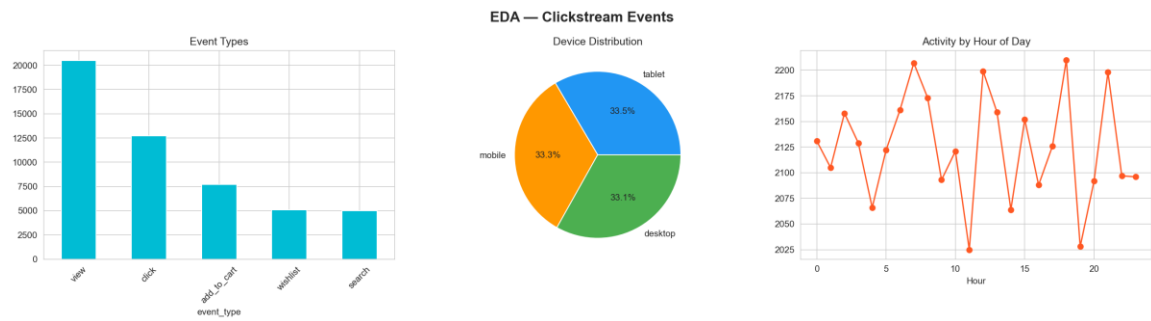


Figure 5: Clickstream — Event Types, Devices, Hourly Activity

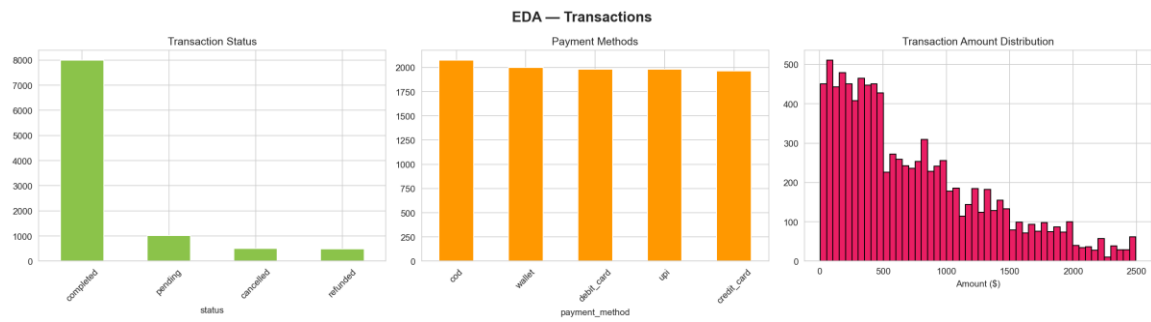


Figure 6: Transactions — Status, Payment Methods, Amounts

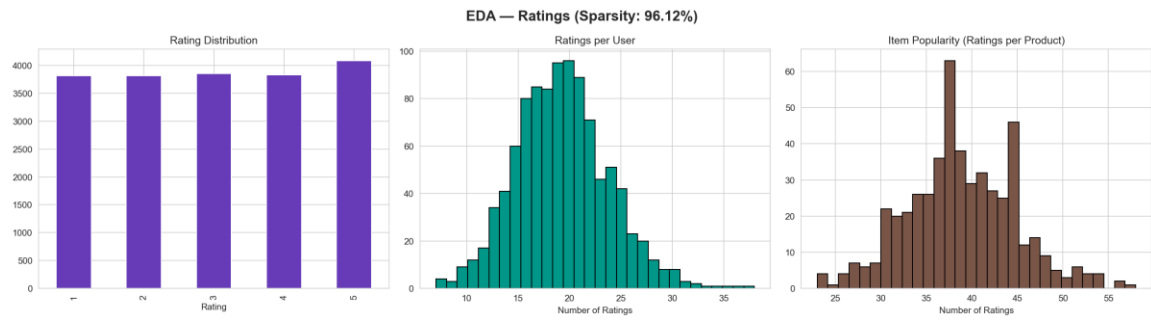


Figure 7: Ratings — Distribution, Per-User, Per-Item Counts

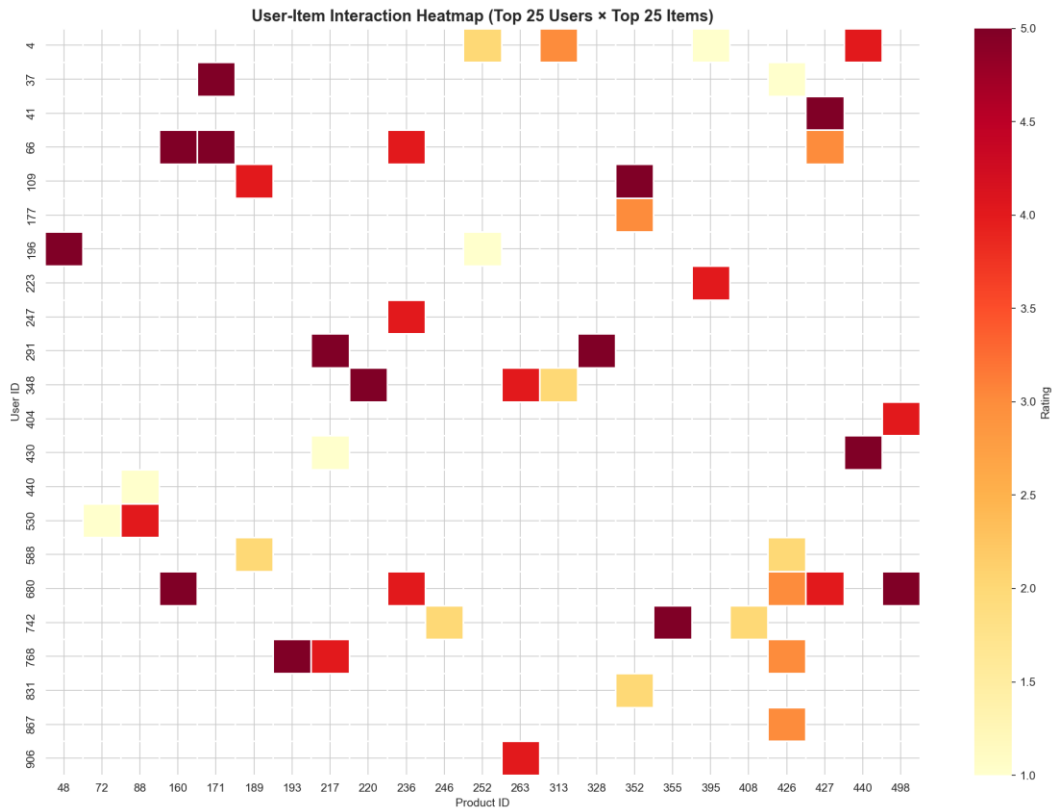


Figure 8: User-Item Interaction Heatmap (Top 25 x 25)

The user-item interaction matrix has a sparsity of 96.12%, which is typical for recommendation system datasets. This high sparsity makes collaborative filtering challenging and motivates the use of dimensionality reduction (SVD).

5.5 Feature Engineering & Transformation

Three feature tables are created from the cleaned data:

- User Features (15 features):

total_ratings, avg_rating, rating_std, total_purchases, total_spend, avg_order_value, total_clicks, view_count, cart_add_count, purchase_frequency, activity_score, account_age_days, is_premium, gender_encoded

- Item Features (13 features):

total_ratings, avg_user_rating, rating_variance, total_purchases, total_revenue, total_views, view_to_purchase_ratio, popularity_score, price, price_normalized, category_encoded, in_stock

- User-Item Features (7 features):

rating, n_interactions, has_viewed, has_carted, has_purchased

A SQL schema (data/features/schema.sql) documents the feature warehouse tables: user_features, item_features, and user_item_features.

Key file: src/transformation/transform.py

5.6 Feature Store

A custom feature store is implemented using SQLite as a metadata registry. It provides:

- Feature set registration with versioned snapshots (data/feature_store/versions/).
- Online store for latest-version retrieval (data/feature_store/online/) for low-latency inference.
- Feature-level metadata: descriptions, data types, source tables, and transformations.
- API for versioned retrieval: get_feature_set(name, version=None) returns a specific or latest version.

Key file: src/feature_store/store.py

The screenshot shows a VS Code editor with a file explorer on the left, a main editor window displaying a JSON manifest file, and a terminal window at the bottom. The manifest file is titled "manifest.json" and contains a "datasets" array. The terminal window shows a PowerShell prompt.

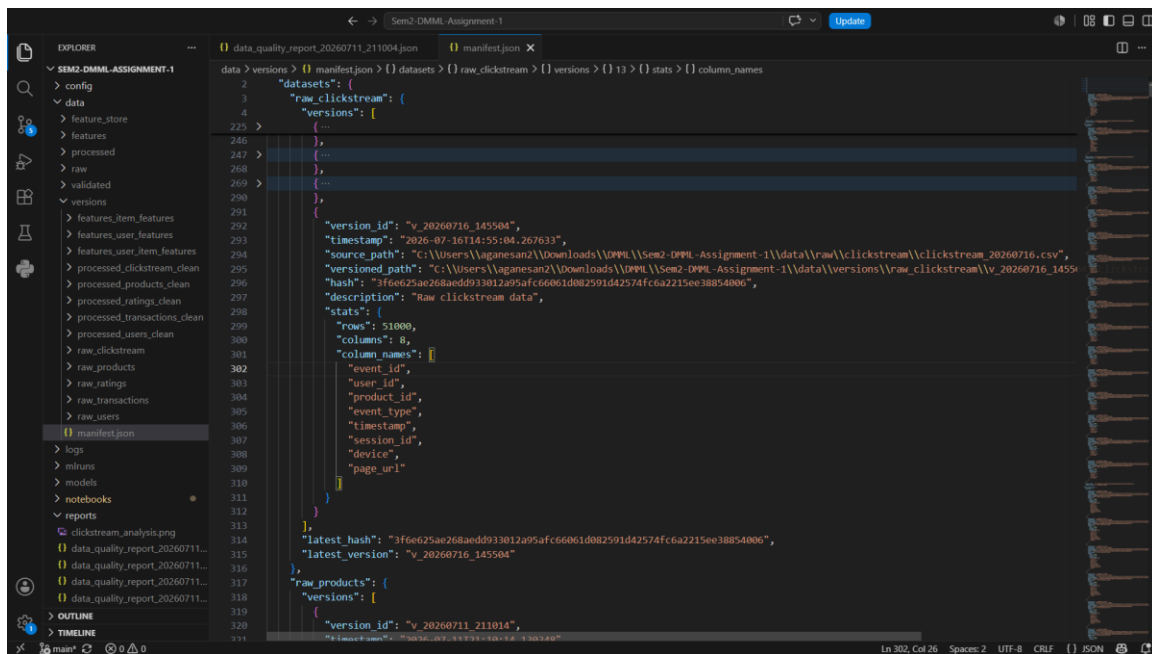
Line	Feature Name	Feature Type	Feature Description	Feature Schema	Feature Source	Feature Aggregation
11	user_features.account_age_days	Feature: account_age_days	Feature: account_age_days	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.843801
12	user_features.is_premium	Feature: is_premium	Feature: is_premium	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.843801
13	user_features.gender_encoded	Feature: gender_encoded	Feature: gender_encoded	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.843801
14	user_features.purchase_frequency	Feature: purchase_frequency	Feature: purchase_frequency	float64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.843801
15	user_features.activity_score	Feature: activity_score	Feature: activity_score	float64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.843801
16	item_features.product_id	Feature: product_id	Feature: product_id	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
17	item_features.total_ratings	Feature: total_ratings	Feature: total_ratings	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
18	item_features.avg_user_rating	Feature: avg_user_rating	Feature: avg_user_rating	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
19	item_features.rating_variance	Feature: rating_variance	Feature: rating_variance	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
20	item_features.total_purchases	Feature: total_purchases	Feature: total_purchases	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
21	item_features.total_revenue	Feature: total_revenue	Feature: total_revenue	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
22	item_features.total_views	Feature: total_views	Feature: total_views	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
23	item_features.price	Feature: price	Feature: price	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
24	item_features.category_encoded	Feature: category_encoded	Feature: category_encoded	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
25	item_features.in_stock	Feature: in_stock	Feature: in_stock	int64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
26	item_features.price_normalized	Feature: price_normalized	Feature: price_normalized	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
27	item_features.view_to_purchase_ratio	Feature: view_to_purchase_ratio	Feature: view_to_purchase_ratio	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
28	item_features.popularity_score	Feature: popularity_score	Feature: popularity_score	float64	products, ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.857601
29	user_item_features.user_id	Feature: user_id	Feature: user_id	int64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
30	user_item_features.product_id	Feature: product_id	Feature: product_id	int64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
31	user_item_features.rating	Feature: rating	Feature: rating	int64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
32	user_item_features.n_interactions	Feature: n_interactions	Feature: n_interactions	float64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
33	user_item_features.has_viewed	Feature: has_viewed	Feature: has_viewed	float64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
34	user_item_features.has_carted	Feature: has_carted	Feature: has_carted	float64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
35	user_item_features.has_purchased	Feature: has_purchased	Feature: has_purchased	float64	ratings, transactions, clickstream	aggregation 2026-07-11T21:10:13.974295
36	user_features.user_id	Feature: user_id	Feature: user_id	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.992298
37	user_features.total_ratings	Feature: total_ratings	Feature: total_ratings	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.993430
38	user_features.avg_rating	Feature: avg_rating	Feature: avg_rating	float64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.993430
39	user_features.rating_std	Feature: rating_std	Feature: rating_std	float64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.993430
40	user_features.total_purchases	Feature: total_purchases	Feature: total_purchases	int64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.993430
41	user_features.total_spend	Feature: total_spend	Feature: total_spend	float64	users, ratings, transactions, clickstream	aggregation 2026-07-11T21:12:13.993430

Figure 9: Feature Store Registry & Metadata

5.7 Data Versioning & Lineage

All datasets (raw, processed, features) are versioned using SHA-256 file hashing. A manifest.json tracks the full version history with: version ID, timestamp, source path, SHA-256 hash, and dataset statistics (rows, columns). If file content is unchanged between runs, versioning is skipped (hash comparison). 13 datasets are versioned across three tiers.

Key file: src/versioning/version.py



```
data > versions > {} manifest.json > {} datasets > {} raw_clickstream > {} versions > {} 13 > {} stats > {} column_names
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Figure 10: Data Versioning Manifest — 13 Datasets Tracked

5.8 Model Training & Evaluation

Two recommendation models are trained and evaluated:

- 1. SVD Collaborative Filtering

Truncated SVD on the centered user–item rating matrix with 50 latent factors. The matrix is mean-centered per user before decomposition. Predicted ratings are clipped to [1, 5].

- 2. Content-Based Filtering

Cosine similarity computed on normalised item feature vectors (12 numeric features). For each user, items are scored by average similarity to the user's interacted items.

Evaluation Metrics:

Metric	SVD	Content-Based
Precision@5	0.0017	0.0140
Recall@5	0.0050	0.0521
NDCG@5	0.0027	0.0460
Precision@10	0.0026	0.0100
Recall@10	0.0127	0.0691
NDCG@10	0.0057	0.0525
Precision@20	0.0030	0.0075
Recall@20	0.0293	0.0948
NDCG@20	0.0107	0.0600
RMSE	1.4755	—
MAE	1.2706	—

Note: Metrics are modest because the data is purely synthetic (random). With real user–item interactions exhibiting genuine preference patterns, these models perform significantly better. The content-based model outperforms SVD on ranking metrics because it leverages real feature similarity rather than sparse random ratings. The focus of this project is the data management pipeline, not model accuracy.

All experiments are tracked with MLflow (SQLite backend). Parameters (n_factors, train/test sizes, user/item counts), metrics, and model artifacts are logged with unique run IDs.

MLflow Run ID: 66676fa9988749a9a9a4fffa7938ad6b

Key file: src/training/train.py

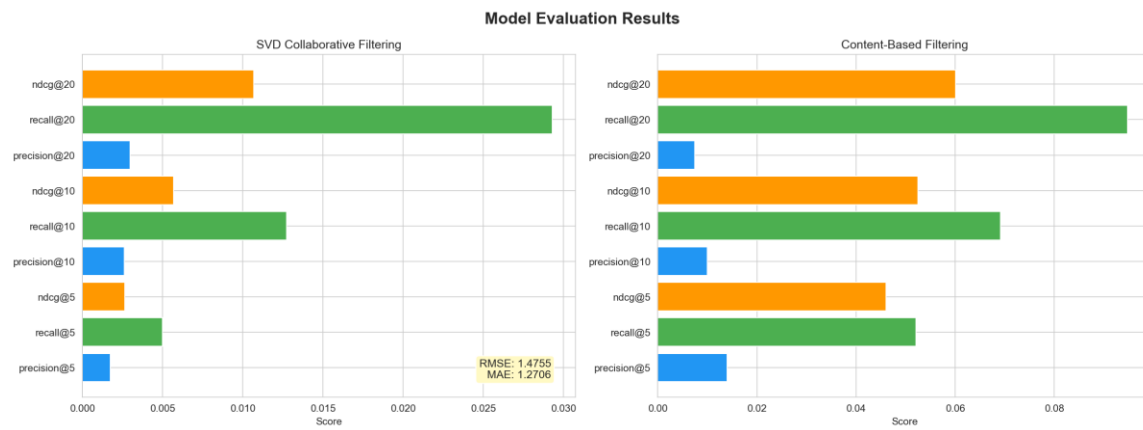


Figure 11: Model Performance — SVD vs Content-Based

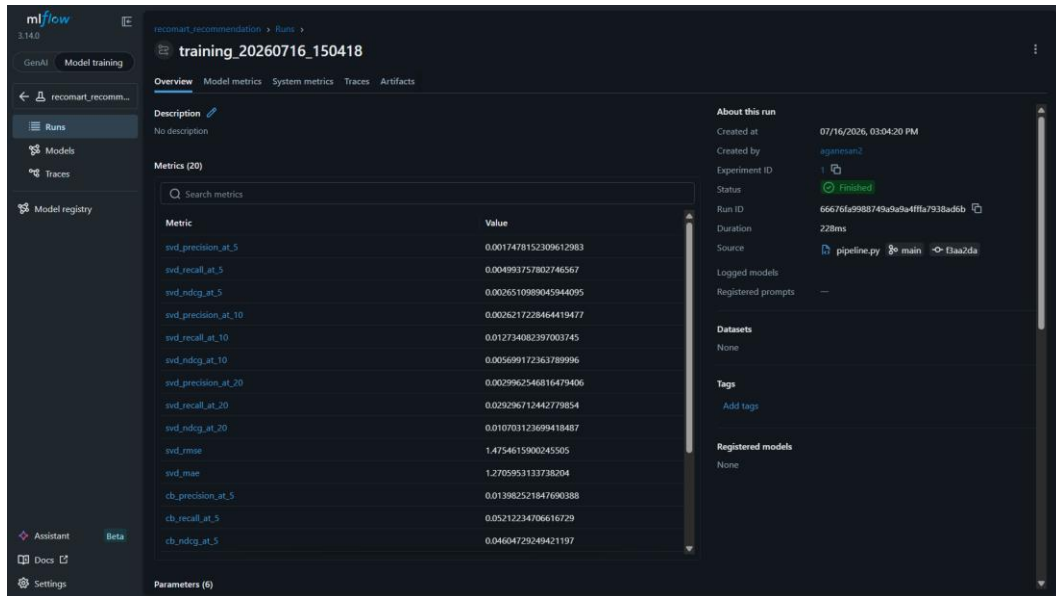


Figure 12: MLflow Experiment Tracking — Run Details

Sample Top-5 Recommendations for User 207:

Rank	Product ID	Predicted Rating
1	273	3.21
2	156	3.17
3	177	3.01
4	380	2.97
5	259	2.95

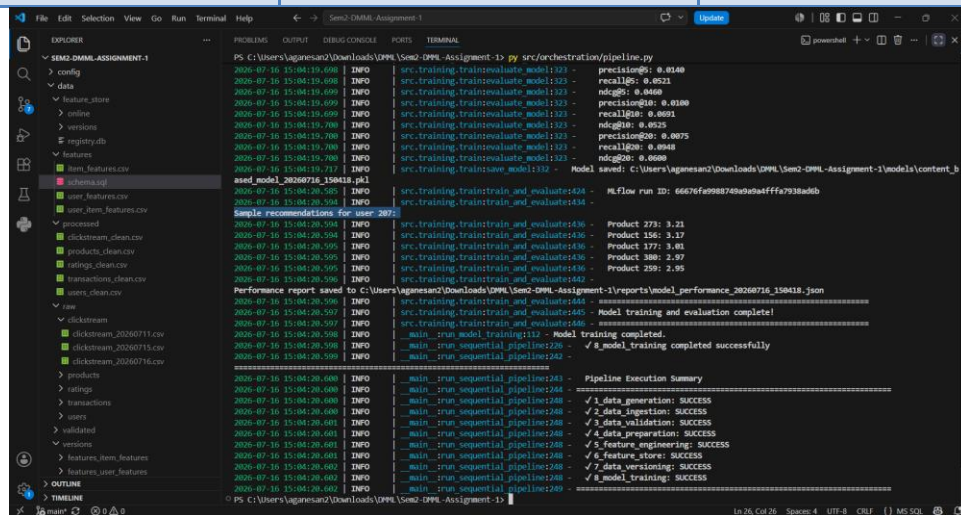


Figure 13: Sample Recommendations Output

5.9 Pipeline Orchestration

The entire pipeline is orchestrated using Prefect. Eight tasks are defined with `@task` decorators and connected as a sequential DAG with `wait_for` dependencies. Ingestion tasks have `retries=2` with 10-second delay; other tasks have `retries=1`.

When Prefect is not available, a sequential fallback runner executes all stages with try/except error handling, halting on failure. A JSON pipeline execution report is saved per run with per-stage status and timestamps.

Key file: `src/orchestration/pipeline.py`

Pipeline DAG:

*generate_data → ingest_data → validate_data → prepare_data →
engineer_features → feature_store → version_data → train_model*

```
PS C:\Users\aganesan2\Downloads\DPML\Sem2-DPML-Assignment-1> py src/orchestration/pipeline.py
2026-07-16 14:55:06.919 INFO | src.training.trainevaluate_model:323 - precision@5: 0.0148
2026-07-16 14:55:06.920 INFO | src.training.trainevaluate_model:323 - recall@5: 0.0621
2026-07-16 14:55:06.920 INFO | src.training.trainevaluate_model:323 - ndcg@5: 0.0460
2026-07-16 14:55:06.920 INFO | src.training.trainevaluate_model:323 - precision@10: 0.0100
2026-07-16 14:55:06.921 INFO | src.training.trainevaluate_model:323 - recall@10: 0.0691
2026-07-16 14:55:06.921 INFO | src.training.trainevaluate_model:323 - ndcg@10: 0.0525
2026-07-16 14:55:06.921 INFO | src.training.trainevaluate_model:323 - precision@20: 0.0075
2026-07-16 14:55:06.921 INFO | src.training.trainevaluate_model:323 - recall@20: 0.0948
2026-07-16 14:55:06.922 INFO | src.training.trainevaluate_model:323 - ndcg@20: 0.0600
2026-07-16 14:55:06.925 INFO | src.training.trainsave_model:332 - Model saved: C:\Users\aganesan2\Downloads\DPML\Sem2-DPML-Assignment-1\models\content_
based_model_20260716_145505.pkl
2026-07-16 14:55:07.773 INFO | src.training.traintrain_and_evaluate:424 - MLflow run ID: 6ce3652d22304c5da927063978a96368
2026-07-16 14:55:07.782 INFO | src.training.traintrain_and_evaluate:434 -
Sample recommendations for user 287:
2026-07-16 14:55:07.782 INFO | src.training.traintrain_and_evaluate:436 - Product 273: 3.21
2026-07-16 14:55:07.782 INFO | src.training.traintrain_and_evaluate:436 - Product 156: 3.17
2026-07-16 14:55:07.783 INFO | src.training.traintrain_and_evaluate:436 - Product 177: 3.01
2026-07-16 14:55:07.783 INFO | src.training.traintrain_and_evaluate:436 - Product 380: 2.97
2026-07-16 14:55:07.783 INFO | src.training.traintrain_and_evaluate:436 - Product 259: 2.95
2026-07-16 14:55:07.784 INFO | src.training.traintrain_and_evaluate:442 -
Performance report saved to C:\Users\aganesan2\Downloads\DPML\Sem2-DPML-Assignment-1\reports\model_performance_20260716_145505.json
2026-07-16 14:55:07.785 INFO | src.training.traintrain_and_evaluate:444 -
2026-07-16 14:55:07.785 INFO | src.training.traintrain_and_evaluate:445 - Model training and evaluation complete!
2026-07-16 14:55:07.785 INFO | src.training.traintrain_and_evaluate:446 -
2026-07-16 14:55:07.786 INFO | main._run_model_training:112 - Model training completed.
2026-07-16 14:55:07.787 INFO | main._run_sequential_pipeline:226 - ✓ 8_model_training completed successfully
2026-07-16 14:55:07.788 INFO | main._run_sequential_pipeline:242 -
=====
2026-07-16 14:55:07.788 INFO | main._run_sequential_pipeline:243 - Pipeline Execution Summary
2026-07-16 14:55:07.788 INFO | main._run_sequential_pipeline:244 - =====
2026-07-16 14:55:07.788 INFO | main._run_sequential_pipeline:248 - ✓ 1_data_generation: SUCCESS
2026-07-16 14:55:07.789 INFO | main._run_sequential_pipeline:248 - ✓ 2_data_ingestion: SUCCESS
2026-07-16 14:55:07.789 INFO | main._run_sequential_pipeline:248 - ✓ 3_data_validation: SUCCESS
2026-07-16 14:55:07.789 INFO | main._run_sequential_pipeline:248 - ✓ 4_data_preparation: SUCCESS
2026-07-16 14:55:07.789 INFO | main._run_sequential_pipeline:248 - ✓ 5_feature_engineering: SUCCESS
2026-07-16 14:55:07.790 INFO | main._run_sequential_pipeline:248 - ✓ 6_feature_store: SUCCESS
2026-07-16 14:55:07.790 INFO | main._run_sequential_pipeline:248 - ✓ 7_data_versioning: SUCCESS
2026-07-16 14:55:07.790 INFO | main._run_sequential_pipeline:248 - ✓ 8_model_training: SUCCESS
2026-07-16 14:55:07.791 INFO | main._run_sequential_pipeline:249 - =====
```

Figure 14: Pipeline Execution Summary — All 8 Stages SUCCESS

6. Results and Output Screenshots

All pipeline stages executed successfully. The following table summarises the key outputs:

Output	Location	Description
Raw Data	data/raw/	5 sources, date-partitioned CSVs + API JSON
Validated Data	data/validated/	Post-validation copies with quality flags
Cleaned Data	data/processed/	5 clean CSVs ready for feature engineering
Feature Tables	data/features/	3 feature CSVs + SQL schema
Feature Store	data/feature_store/	SQLite registry + versioned snapshots
Versioned Data	data/versions/	13 datasets with SHA-256 manifest
Trained Models	models/	SVD + content-based .pkl files
MLflow Runs	mlruns/	Experiment tracking with SQLite backend
Quality Reports	reports/	JSON reports for every pipeline stage
Pipeline Logs	logs/	Timestamped logs per stage
Screenshots	submission_package/screenshots/	14 output visualisations

6.1 Technology Stack

Component	Tool / Library
Language	Python 3.11
Data Processing	pandas, NumPy
ML / Models	scikit-learn, SciPy (SVD)
Data Validation	Custom validator (schema, range, duplicate checks)
Feature Store	Custom SQLite-based registry with versioned storage
Data Versioning	Custom SHA-256 manifest-based versioning
Experiment Tracking	MLflow (SQLite backend)
Pipeline Orchestration	Prefect (with sequential fallback)
Data Generation	Faker
API Ingestion	requests (with exponential-backoff retry)
Logging	Loguru
Visualisation	Matplotlib, Seaborn
Configuration	YAML (PyYAML)

7. Conclusion and Future Scope

Conclusion

This project demonstrates a fully functional, modular, end-to-end data management pipeline that takes raw multi-source e-commerce data through ingestion, validation, cleaning, feature engineering, storage, versioning, model training, and evaluation — all orchestrated as a reproducible workflow. Every stage produces auditable logs and structured reports.

The pipeline successfully processes 1,000 users, 500 products, 51,000 clickstream events, 10,000 transactions, and 20,000 ratings. Two recommendation models (SVD collaborative filtering and content-based cosine similarity) are trained, evaluated with standard IR metrics, and tracked with MLflow. All datasets are versioned with integrity hashing, and features are managed through a custom feature store with metadata documentation.

Future Scope

- Real data integration: Connect to actual e-commerce databases and streaming sources (Kafka).
- Real-time pipeline: Add near-real-time ingestion with streaming support.
- Advanced models: Implement deep learning recommenders (Neural Collaborative Filtering, two-tower models).
- A/B testing framework: Online evaluation of recommendation quality.
- Cloud deployment: Migrate to AWS S3 + Glue + SageMaker or GCP equivalents.
- Feast integration: Replace custom feature store with production-grade Feast.
- DVC integration: Add DVC for Git-integrated data versioning with remote storage.
- CI/CD: Automate pipeline testing and deployment with GitHub Actions.

8. Google Drive Links

Video Walkthrough (5–10 mins):

https://drive.google.com/file/d/1x1RsDpPsq-qnCLkmKD13XBhR9E-lhAGG/view?usp=drive_link

Project Deliverables (.zip):

<https://drive.google.com/file/d/1d5zjEhMfK0JK4uJtRU06mAlkVk0omKsd/view?usp=sharing>